

(Sec 2–3) Inference-Time Availability Threats — a unified view

Unifying principle: Intermediate-Work Amplification (force more boxes / tokens / steps through super -linear-complexity stages)

Attacker interface

Knowledge (white/grey/black)

Targeted stage

Exploited bottleneck

Metric (latency/energy/FLOPs)

(Sec 4–5) Attacks by Target

4 Object detection & perception

- 4.1 NMS bottleneck
- 4.2 pipeline/MOT
- 4.3 & 4.4 physical-world/LiDAR
- 4.5 embodied & VLA

5 Neural inference work-inflation

- 5.1 sponge
- 5.2 poisoning
- 5.3 sparsity/spiking
- 5.4 dynamic & adaptive networks

(Sec 6–7) Generative & Persistent Attacks

6 Autoregressive & serving-side

- 6.1 sequence generation (NMT/caption/TTS)
- 6.2 LLM & VLM resource attacks
 - output-length inflation / DoS
 - reasoning 'infinite-thinking'
 - multimodal verbose attacks

7 Training-time / HW / timing channels

(Sec 8) Defenses by Mechanism

8.1 Budget enforcement

8.2 Robust / adaptive training

8.3 Input transformation

8.4 Bounded / constant-time execution

8.5 Architectural sparsity

8.6 Runtime monitoring

8.7 System / serving control

(Sec 9–11) Synthesis & Open Directions

9 Cross-domain synthesis & evaluation

- evaluation practices & benchmarks
- work-budget abstraction

10 Emerging surfaces & open challenges

- 10.1 NMS-free detectors
- 10.2 cross-cutting open challenges

11 Conclusion